

Lab 4: Putting it all together

Hadley Stabnau

2026-02-05

Reading in data

```
library(tidyverse)

relig_income <- read_csv(
  "~/Desktop/STAT408/relig_income.csv",
  skip = 2
)
```

In order to read in this data, I had to first load the tidyverse package in order to be able to use the `read_csv()` function. Next, I read in the data, using an absolute file path. Then, I had to skip the first two lines of the data from `relig_income.csv`, as they are not actually part of the data.

Exploring the data

```
[1] 11
```

```
[1] 18
```

```
[1] "Religion of respondent" "<$10k" "$10-20k"
[4] "$20-30k" "$30-40k" "$40-50k"
[7] "$50-75k" "$75-100k" "$100-150k"
[10] ">150k" "Don't know/refused"
```

- There are 11 columns/variables in the religios income dataset, these are: Religion of respondent, <\$10k, \$10-20k, \$20-\$30k, \$30-\$40k, \$40-\$50k, \$50-\$75k, \$75-\$100k, \$100-\$150k, >\$150k, and Don't know/refused.
- There are 18 rows of this dataset, indicating 18 different religions included.
- The first column/variable contains qualitative values that are of the categorical variable: religion. The other columns have quantitative values. None of the variables have missing values and at first glance, I think some kind of bar graph would be useful for visualizing.

Summary Statistics

```
relig_income |>
  select(`Religion of respondent`, `<$10k`) |>
  arrange(desc(`<$10k`))
```

```
# A tibble: 18 x 2
  `Religion of respondent` `<$10k`
  <chr>                    <dbl>
1 Evangelical Prot        575
2 Catholic                418
3 Mainline Prot           289
4 Historically Black Prot  228
5 Unaffiliated            217
6 Mormon                  29
7 Agnostic                27
8 Buddhist                27
9 Jehovah's Witness       20
10 Other Faiths            20
11 Jewish                  19
12 Don't know/refused      15
13 Orthodox                13
14 Atheist                 12
15 Other Christian          9
16 Muslim                   6
17 Other World Religions    5
18 Hindu                    1
```

- The religion with the most amount of people in the less than \$10 thousand dollar income group is the evangelical protestants.

```
relig_income |>
  select(`Religion of respondent`, `>150k`) |>
  arrange(desc(`>150k`))
```

```
# A tibble: 18 x 2
  `Religion of respondent` `>150k`
  <chr>                   <dbl>
1 Mainline Prot           634
2 Catholic                633
3 Evangelical Prot       414
4 Unaffiliated           258
5 Jewish                  151
6 Agnostic                 84
7 Historically Black Prot  78
8 Atheist                  74
9 Hindu                    54
10 Buddhist                53
11 Orthodox                46
12 Mormon                  42
13 Other Faiths            41
14 Don't know/refused      18
15 Other Christian         12
16 Jehovah's Witness        6
17 Muslim                   6
18 Other World Religions    4
```

- The religion with the most amount of people in the greater than 150 thousand dollar income group is the mainline protestants, followed very closely by the Catholics.

Plot the Data

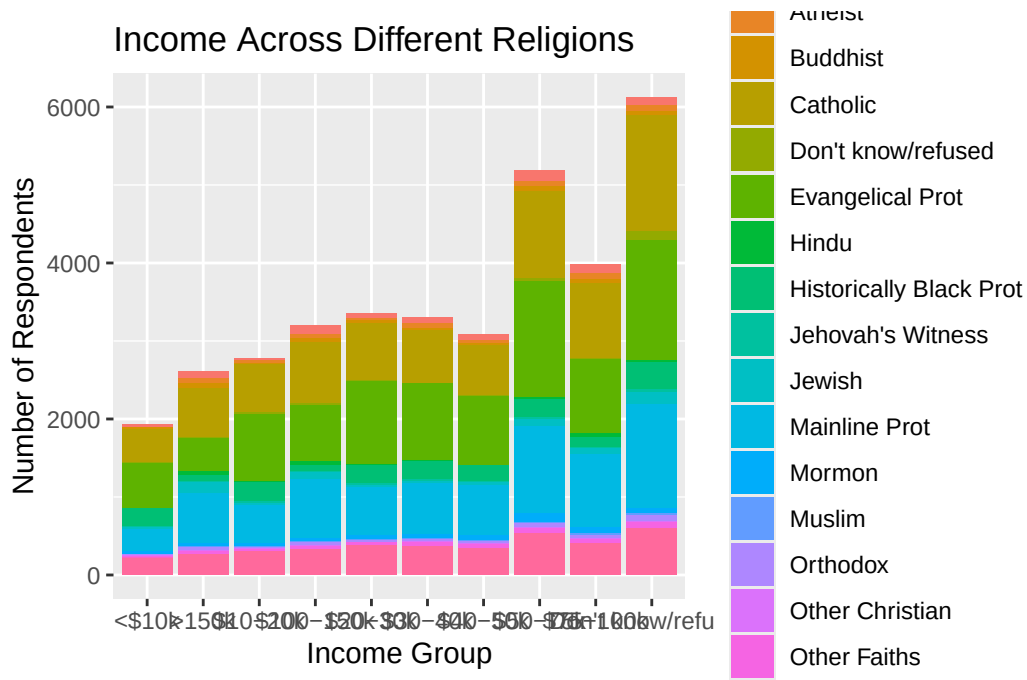
```
library(tidyverse)

relig_long = relig_income |>
  pivot_longer(
    cols = -`Religion of respondent`,
    names_to = "Income Group",
    values_to = "Count"
  )
```

```

ggplot(data = relig_long,
       aes(
         x = `Income Group`,
         y = Count,
         fill = `Religion of respondent`
       )) +
  geom_col() +
  labs(
    x = "Income Group",
    y = "Number of Respondents",
    fill = "Religion",
    title = "Income Across Different Religions"
  )

```



- I could not quite figure out how to orient the labels in a way so that they aren't so squished you can't read them. But the rest of the graph is organized and labeled. The different bars are representing each income group and the different colors of the bars represent the religions that make up each of the bars.

Analysis

- This data could be examined further to look at the number of responses across the different income groups. Some of the bars are much taller than others, it makes me wonder what kind of bias/obstacles lead to more data in some groups than others. I mentioned before that I don't feel like I yet have the skills to further organize the labels of income groups across the x axis. Another thing that I would like to do with this data if I had the skills, is use R to look at the percentages of each religion in each income group and compare them. I earlier looked at counts, but it would be cool to use R to simply pull up the percent of each religion in each group all in one table for analysis.